# DevOps interview

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IDEA



“We deployed a WordPress application with auto-scaling capabilities on AWS, using Terraform for infrastructure provisioning”

## OBJECTIVES

### Infrastructure as Code

Leverages Terraform and Kubernetes manifests to provide a fully automated, version-controlled deployment lifecycle on AWS.

### Scalability

Employs autoscaling to dynamically adjust application replicas based on real-time load. Utilizing the EKS Nodes to ensure the underlying compute capacity can be seamlessly scaled.

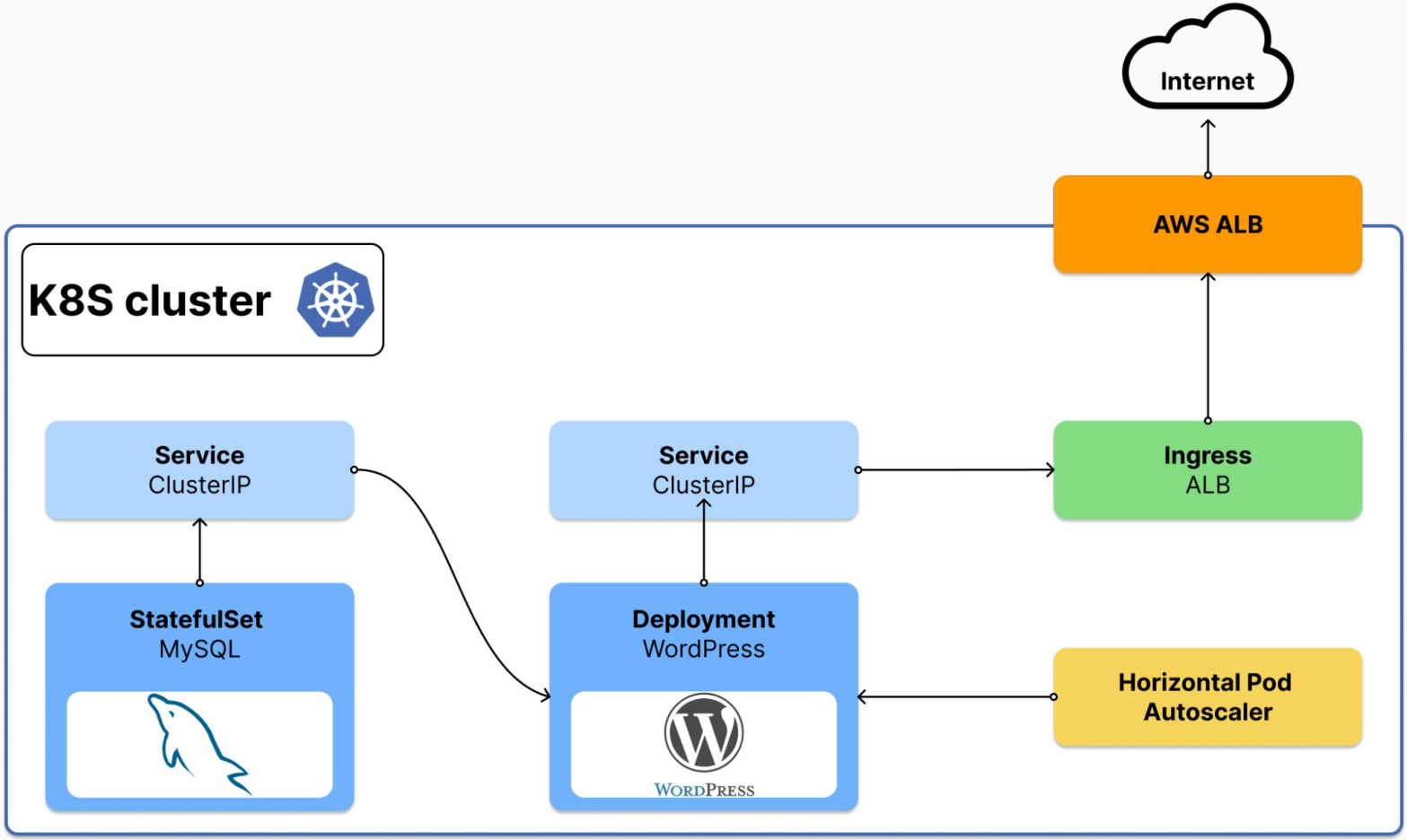
### Modularity

Utilizes a decoupled architecture where infrastructure is separated into reusable Terraform modules and application resources into Kustomize components.

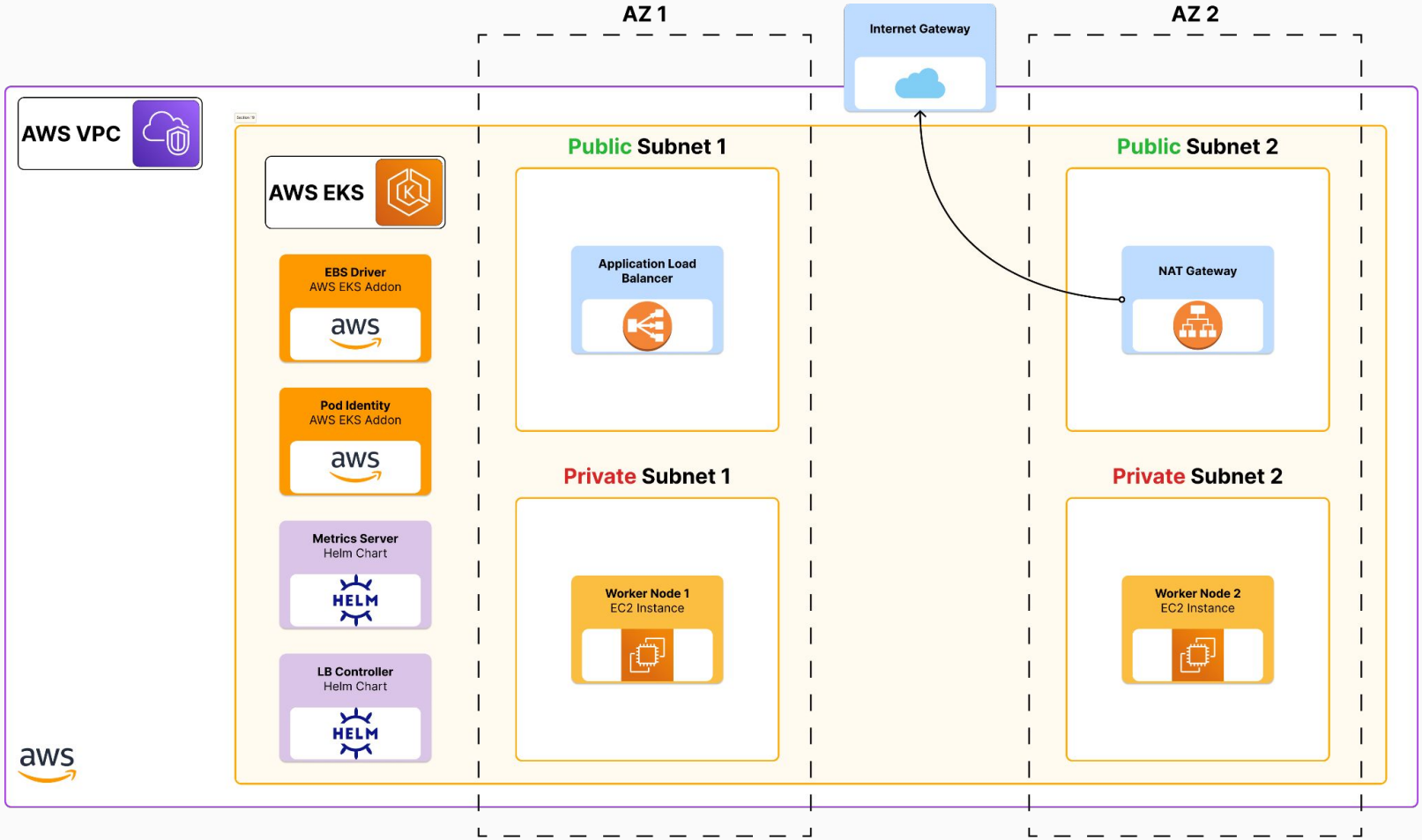
### Resilience

Ensures high availability and data integrity through Multi-AZ networking, Kubernetes self-healing, and persistent storage using the Amazon EBS.

## APPLICATION OVERVIEW



# INFRASTRUCTURE OVERVIEW



**Modules**

## Kubernete Overlay (IaC)

- Employs a Kustomize "Base and Overlay" pattern across components and overlays.
- It defines a modular wordpress-app bundle featuring a stateless Nginx frontend, a MySQL StatefulSet for stable identity, and a gp3 StorageClass for dynamic persistence.
- Overlays manage environment specific patches and automated Secret Generation.

## Terraform (IaC)

- Organized into modules for reusable core logic (Network, EKS, Addons) and infra/env/ for environment-specific states.
- It provisions a dev environment EKS cluster (v1.35) using cost-efficient SPOT instances and automates the deployment of essential AWS integration addons like the Load Balancer Controller and EBS CSI driver.

## GitHub Actions (CI)

- Acts as the project's automated quality gate, running continuous integration workflows for every change.
- It performs Terraform formatting check and plan validation, Kustomize build checks, and YAML linting to ensure that only high-quality, valid code is deployed to the infrastructure.

## PROJECT STRUCTURE

### Kubernete (IaC)

```
k8s
├── components
│   └── wordpress-app
│       ├── database
│       ├── kustomization.yaml
│       ├── ns.yaml
│       ├── storage
│       └── wordpress
└── overlays
    └── devops-demo-eks-aws-us-east-1
        ├── kustomization.yaml
        ├── mysql-secrets.env
        ├── mysql-secrets-example.env
        ├── wordpress-secrets.env
        └── wordpress-secrets-example.env
```

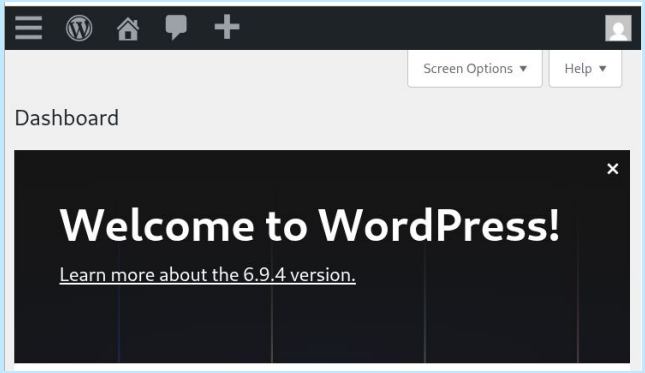
### Terraform (IaC)

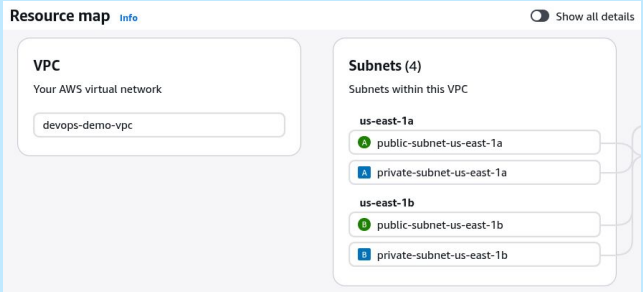
```
infra
├── env
│   └── dev
│       ├── backend.tf
│       ├── main.tf
│       ├── outputs.tf
│       ├── providers.tf
│       ├── terraform.tfvars
│       ├── variables.tf
│       └── versions.tf
└── modules
    ├── addons
    │   ├── ebs_csi
    │   ├── load_balancer
    │   ├── metrics_server
    │   └── pod_identity
    ├── eks
    │   ├── main.tf
    │   ├── outputs.tf
    │   └── variables.tf
    └── network
        ├── main.tf
        ├── outputs.tf
        └── variables.tf
```

### GitHub Actions (CI)

```
.github
└── workflows
    ├── kustomization-validaton.yaml
    ├── terraform-validation.yaml
    ├── yaml-lint.yaml
    └── yamllint.yaml
```

# WordPress application




```
Every 1.0s: kubectl get pods -l app=wordpress -A

NAMESPACE                NAME                         READY   STATUS    RESTARTS        AGE
devops-demo--wordpress   wordpress-748999b48c-2ttvn   1/1     Running   0               2m12s
devops-demo--wordpress   wordpress-748999b48c-6b946   0/1     Pending   0               27s
devops-demo--wordpress   wordpress-748999b48c-6dzts   0/1     Pending   0               27s
devops-demo--wordpress   wordpress-748999b48c-8wmq4   1/1     Running   0               2m27s
devops-demo--wordpress   wordpress-748999b48c-hvfsg   1/1     Running   0               2m27s
devops-demo--wordpress   wordpress-748999b48c-ssq2d   1/1     Running   1 (6m31s ago)   7m33s
devops-demo--wordpress   wordpress-748999b48c-vksfs   1/1     Running   0               2m12s
devops-demo--wordpress   wordpress-748999b48c-x4r7j   0/1     Pending   0               2m12s
```

```
Every 1.0s: kubectl get hpa wordpress -n devops-demo--wordpress

NAME        REFERENCE              TARGETS                        MINPODS   MAXPODS   REPLICA
wordpress   Deployment/wordpress   cpu: 136%/60%, memory: 53%/90%   1         8         8
```

```
Every 1.0s: kubectl top nodes

NAME                          CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
ip-10-0-0-159.ec2.internal    1190m        61%      1203Mi          16%
ip-10-0-1-57.ec2.internal     2001m        103%     1047Mi          33%
```

# Thank you!

Artur Sultanov